



## BayesianDEB: A Bayesian Framework for Dynamic Energy Budget Modelling in R

Branimir K. Hackenberger  
University of Osijek, SCIOM

Tamara Djerdj  
University of Osijek

Domagoj K. Hackenberger  
Ruder Bošković Institute

---

### Abstract

Dynamic Energy Budget (DEB) theory provides a mechanistic, thermodynamically consistent description of how organisms acquire and utilise energy for maintenance, growth, development, and reproduction. Classical DEB calibration relies on deterministic optimisation, which yields no formal uncertainty quantification and cannot naturally accommodate prior biological knowledge or hierarchical data structures. The R package **BayesianDEB** addresses these limitations by embedding the DEB ordinary differential equation system within a Bayesian state-space framework, using **Stan** for Hamiltonian Monte Carlo inference. The package provides four pre-written **Stan** programs covering common DEB applications: individual growth, growth with reproduction, hierarchical multi-individual analysis with partial pooling, and toxicokinetic-toxicodynamic (TKTD) models for ecotoxicology. A declarative R interface handles data preparation, prior specification from biological knowledge, observation model selection, Arrhenius temperature correction, MCMC fitting with within-chain parallelism, convergence diagnostics, posterior predictive checks, derived quantity estimation, and LOO cross-validation. We demonstrate the full workflow on simulated *Eisenia fetida* growth data, on real growth data from the literature, and on a DEBtox analysis with  $EC_{50}$ /NEC estimation. The individual growth model is validated via simulation-based calibration with 500 replications. An identifiability analysis shows that DEB rate parameters are largely prior-dominated under single-individual growth data, underscoring the importance of species-specific priors or hierarchical modelling for quantitative inference. The package is available from the SCIOM r-universe at <https://sciom.r-universe.dev/BayesianDEB>; CRAN submission is under review.

*Keywords:* dynamic energy budget, Bayesian inference, **Stan**, ecotoxicology, DEBtox, TKTD, hierarchical models, R.

---

## 1. Introduction

Dynamic Energy Budget (DEB) theory (Kooijman 2010) provides a mechanistic, thermo-

dynamically grounded framework for describing energy acquisition, storage, and allocation across the life cycle of an organism, and is among the most widely applied metabolic theories in ecology and ecotoxicology. Its core state variables — reserve energy  $E$ , structural volume  $V$ , maturity  $E_H$ , and reproduction buffer  $E_R$  — evolve according to an ordinary differential equation (ODE) system governed by the  $\kappa$ -rule: a fraction  $\kappa$  of mobilised reserve is allocated to somatic maintenance and growth, while the remainder funds maturity maintenance and reproduction.

Despite its theoretical elegance, DEB calibration has traditionally relied on deterministic least-squares or maximum-likelihood optimisation, as implemented in **DEBtool\_M** (Marques *et al.* 2018) and the Add-my-Pet (AmP) online estimation tool. This approach yields point estimates without formal uncertainty quantification, cannot incorporate prior biological knowledge in a principled way, and does not naturally extend to hierarchical data structures where multiple individuals share species-level parameters but differ in, for example, assimilation rate.

The combination of Bayesian inference with ODE-based models is, of course, not new (Gelman *et al.* 2013); several authors have applied custom MCMC to DEB (Johnson *et al.* 2015), and **deBInfer** (Boersch-Supan *et al.* 2017) provides a general Bayesian ODE framework in R. The contribution of the present work is not a new statistical method but a *domain-specific software system* that makes three concrete advances over existing tools:

1. *Unified Bayesian workflow for DEB.* No existing tool combines DEB ODE models, HMC inference, observation model switching, Arrhenius temperature correction, convergence diagnostics, posterior predictive checks, derived quantity estimation, and LOO cross-validation in a single R interface that requires no recompilation when switching observation families or prior specifications. Users of **DEBtool\_M** or custom **Stan** code must assemble this workflow manually; **BayesianDEB** provides it out of the box for the four most common DEB applications.
2. *Reproducible, species-specific prior system grounded in the AmP collection.* The function `prior_species()` returns priors calibrated to AmP parameter estimates (Marques *et al.* 2018) for common test species, bridging the gap between the world’s largest DEB parameter database and Bayesian inference. The prior-to-posterior contraction diagnostic (Section 6.2) quantifies how much the data inform each parameter beyond the prior — a transparency that deterministic calibration cannot offer.
3. *Inference validation for the individual growth model.* Simulation-based calibration (Section 6.1) with 500 replications and formal  $\chi^2$  uniformity tests demonstrates, conditional on MCMC convergence, that the individual growth model is well-calibrated. Preliminary SBC evidence for the hierarchical model is also provided. To our knowledge, this level of algorithmic validation has not been reported for any DEB calibration tool.

Together, these advances lower the barrier to rigorous Bayesian DEB modelling from expert **Stan** programming to a declarative R call, while making identifiability limitations and prior sensitivity explicit rather than hidden.

*Important caveat.* As our identifiability analysis (Section 6.2) demonstrates, growth-only data from a single individual cannot resolve the individual contributions of the DEB rate parameters  $\{p_{Am}\}$ ,  $[p_M]$ ,  $v$ ; posteriors for these parameters are substantially shaped by the

prior even with dense time series. Species-specific informative priors or hierarchical data structures are essential for quantitative inference. **BayesianDEB** makes this limitation *visible* through contraction diagnostics and prior sensitivity tools, whereas deterministic calibration tools obscure it behind point estimates.

The **BayesianDEB** package wraps four **Stan** (Carpenter *et al.* 2017) programs with a declarative R interface. The package implements:

- Individual-level growth (2-state:  $E, V$ ).
- Growth with reproduction (3-state:  $E, V, E_R$ ).
- Hierarchical multi-individual growth with partial pooling on  $\{p_{Am}\}$  via a non-centred parameterisation (Betancourt and Girolami 2015).
- DEBtox toxicokinetic-toxicodynamic (TKTD) models following Jager *et al.* (2006), with analytical  $EC_{50}$  and NEC computation.

All models support switchable observation families (Gaussian, log-normal, Student- $t$  for growth; negative binomial or Poisson for reproduction), Arrhenius temperature correction, within-chain parallelism via **Stan**'s `reduce_sum`, and a suite of diagnostic and visualisation tools.

This paper describes **BayesianDEB** version 0.1.2. The package is available from the SCIOm r-universe at <https://sciom.r-universe.dev/BayesianDEB> (CRAN submission under review) and from GitHub at <https://github.com/sciom/BayesianDEB>.

## 2. DEB theory and model formulation

We summarise the DEB equations implemented in **BayesianDEB** and discuss modelling choices; for a full treatment see Kooijman (2010). The standard DEB model is one of several metabolic frameworks; alternatives include the “standard” and “abj” typified models in the AmP collection (Marques *et al.* 2018), net-production models, and various simplified formulations (Jager and Zimmer 2012). **BayesianDEB** implements the standard ( $\kappa$ -rule) formulation, which is the most widely used and has the largest empirical parameter base via AmP.

### 2.1. Standard DEB: growth

The two core state variables are reserve energy  $E$  (J) and structural volume  $V$  ( $\text{cm}^3$ ), with structural length  $L = V^{1/3}$ . Under constant food availability characterised by the scaled functional response  $f \in [0, 1]$ , the energy fluxes are:

$$\dot{p}_A = f \{p_{Am}\} L^2, \quad (1)$$

$$\dot{p}_C = \frac{E v L}{E + [E_G] V}, \quad (2)$$

$$\dot{p}_M = [p_M] V, \quad (3)$$

where  $\{p_{Am}\}$  is the surface-area-specific maximum assimilation rate ( $\text{J d}^{-1} \text{cm}^{-2}$ ),  $v$  is the energy conductance ( $\text{cm d}^{-1}$ ),  $[E_G]$  is the specific cost of structure ( $\text{J cm}^{-3}$ ), and  $[p_M]$  is the

volume-specific somatic maintenance rate ( $\text{J d}^{-1} \text{cm}^{-3}$ ). The state equations are:

$$\frac{dE}{dt} = \dot{p}_A - \dot{p}_C, \quad (4)$$

$$\frac{dV}{dt} = \frac{\kappa \dot{p}_C - \dot{p}_M}{[E_G]}, \quad (5)$$

where  $\kappa$  is the allocation fraction to soma. At steady state ( $dV/dt = 0$ ,  $dE/dt = 0$ ), the ultimate structural length is  $L_\infty = f \kappa \{p_{Am}\}/[p_M]$ .

**Edge cases.** When  $\kappa \dot{p}_C < \dot{p}_M$  (starvation), growth ceases and the organism draws on structure for maintenance; in our implementation  $dV/dt$  is clamped to zero to prevent negative structural volume. When  $\kappa \rightarrow 0$ , nearly all mobilised energy goes to reproduction and growth is negligible; when  $\kappa \rightarrow 1$ , reproduction ceases. The Beta prior on  $\kappa$  (default Beta(2, 2)) regularises the posterior away from these boundary cases.

**ODE stiffness and solver choice.** DEB systems are mildly stiff due to the time-scale separation between fast reserve dynamics and slow structural growth. The mobilisation flux  $\dot{p}_C = EvL/(E + [E_G]V)$  creates stiffness when  $E \gg [E_G]V$  (near steady-state reserve). We use the BDF (backward differentiation formula) solver `ode_bdf_tol` with tolerances  $10^{-6}$ , which is standard for mildly stiff biological ODE systems. The explicit Runge-Kutta solver (`ode_rk45`) was also tested: it produced identical trajectories for well-conditioned parameter regions but required approximately  $2\times$  more function evaluations and occasionally failed for extreme parameter combinations during MCMC warmup. The BDF solver was therefore chosen as the safer default.

## 2.2. Growth with reproduction

For post-pubertal organisms ( $E_H \geq E_H^p$ ), adding the reproduction buffer  $E_R$  (J) yields a 3-state model:

$$\frac{dE_R}{dt} = (1 - \kappa) \dot{p}_C - k_J E_H^p, \quad (6)$$

where  $k_J$  is the maturity maintenance rate coefficient and  $E_H^p$  is maturity at puberty (treated as a known constant, not a state variable, since the model describes post-pubertal growth only). Offspring counts over an interval  $[t_s, t_e]$  are modelled as  $R \sim \text{NegBin}_2(\mu = k_R \Delta E_R, \phi)$ , where  $\Delta E_R = E_R(t_e) - E_R(t_s)$ ,  $k_R$  is the buffer-to-offspring conversion rate (offspring  $\text{J}^{-1}$ ),  $\mu$  is the expected count, and  $\phi > 0$  is the overdispersion parameter. This uses **Stan's** `neg_binomial_2` parametrisation, where  $E[R] = \mu$  and  $\text{Var}[R] = \mu + \mu^2/\phi$ ; as  $\phi \rightarrow \infty$  the distribution converges to  $\text{Poisson}(\mu)$ .

## 2.3. DEBtox: toxicokinetic-toxicodynamic extension

The DEBtox framework (Jager *et al.* 2006; Jager and Zimmer 2012) adds a scaled internal damage variable  $D_w$ :

$$\frac{dD_w}{dt} = k_d (\max(C_w - z_w, 0) - D_w), \quad (7)$$

where  $k_d$  is the damage recovery rate,  $C_w$  is the external concentration, and  $z_w$  is the no-effect concentration (NEC). The damage causes a stress factor  $s = b_w D_w$  that reduces assimilation:

$\dot{p}_A \leftarrow \dot{p}_A \cdot \max(1 - s, 0)$ . At steady state,  $D_w = C_w - z_w$  for  $C_w > z_w$ , so the  $\text{EC}_{50}$  for 50% assimilation reduction is:

$$\text{EC}_{50} = z_w + \frac{0.5}{b_w}. \quad (8)$$

## 2.4. Arrhenius temperature correction

All rate parameters scale with temperature by the factor  $c_T = \exp(T_A/T_{\text{ref}} - T_A/T)$  (Kooijman 2010, Eq. 1.2), where  $T_A$  is the Arrhenius temperature (species-specific, typically 6000–12 000 K). In **BayesianDEB**, the correction is applied inside the **Stan** ODE to  $\{p_{Am}\}$ ,  $[p_M]$ ,  $v$ ,  $k_J$ , and  $k_d$ ; the estimated parameters correspond to the reference temperature  $T_{\text{ref}}$ .

## 2.5. Hierarchical structure

When data from multiple individuals are available, a lognormal random effect on  $\{p_{Am}\}$  is introduced via the non-centred parameterisation (Betancourt and Girolami 2015):

$$\log\{p_{Am}\}_j = \mu_{\log\{p_{Am}\}} + \sigma_{\log\{p_{Am}\}} \cdot z_j, \quad z_j \sim \mathcal{N}(0, 1). \quad (9)$$

The shared parameters  $[p_M]$ ,  $\kappa$ ,  $v$ ,  $[E_G]$  are estimated from all individuals jointly (partial pooling).

**Why random effects on  $\{p_{Am}\}$  only?** In DEB theory,  $\{p_{Am}\}$  is the parameter most strongly linked to individual phenotypic variation: it determines food uptake capacity and scales with structural surface area, which varies among individuals due to differences in condition, microhabitat, and genotype. The energy conductance  $v$ , maintenance  $[p_M]$ , and costs  $[E_G]$  are typically more conserved within a species (Kooijman 2010; Marques *et al.* 2018). Placing random effects on all parameters simultaneously would create a high-dimensional, poorly identified model for typical DEB dataset sizes (5–30 individuals with 5–15 time points each). The single-random-effect design is a deliberate parsimony choice; extension to multivariate random effects on  $(\{p_{Am}\}, \kappa)$  or  $(\{p_{Am}\}, L_0)$  is planned for a future version once identifiability under realistic sample sizes has been formally assessed.

## 3. Bayesian state-space formulation

We now formalise the statistical model. Let  $\mathbf{x}(t) = (E(t), V(t))^\top$  (or  $(E, V, E_R)^\top$ ,  $(E, V, E_R, D_w)^\top$  for the extended models) denote the latent DEB state, and let  $\boldsymbol{\theta}$  collect the DEB parameters.

### 3.1. Process model

The latent state evolves deterministically according to the ODE:

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, \boldsymbol{\theta}, f, C_w), \quad \mathbf{x}(0) = \mathbf{x}_0(\boldsymbol{\theta}), \quad (10)$$

where  $\mathbf{f}$  is defined by Equations (4)–(5) (growth), (6) (reproduction), or (7) (DEBtox), and  $\mathbf{x}_0$  depends on initial conditions  $(E_0, L_0)$ .

### 3.2. Observation model

Given the ODE solution  $\hat{L}_i = [\mathbf{x}(t_i)]_V^{1/3}$ , the growth observations are conditionally independent:

$$L_i^{\text{obs}} \mid \hat{L}_i, \sigma_L \sim \begin{cases} \mathcal{N}(\hat{L}_i, \sigma_L) & \text{(Gaussian),} \\ \text{LogNormal}(\log \hat{L}_i, \sigma_L) & \text{(log-normal),} \\ t_\nu(\hat{L}_i, \sigma_L) & \text{(Student-}t\text{).} \end{cases} \quad (11)$$

For reproduction, interval offspring counts use Stan's mean-overdispersion parametrisation (`neg_binomial_2`):

$$R_k \mid \mu_k, \phi \sim \text{NegBin}_2(\mu_k, \phi) \quad \text{or} \quad \text{Poisson}(\mu_k), \quad \mu_k = k_R \Delta E_{R,k}, \quad (12)$$

where  $E[R_k] = \mu_k$  and  $\text{Var}[R_k] = \mu_k + \mu_k^2/\phi$ .

### 3.3. Prior model

The full prior factorises as:

$$p(\boldsymbol{\theta}) = p(\{p_{Am}\}) p([p_M]) p(\kappa) p(v) p([E_G]) p(E_0) p(L_0) p(\sigma_L) [\text{model-specific}]. \quad (13)$$

Default prior families are:  $\{p_{Am}\}, [p_M], v, [E_G] \sim \text{LogNormal}$ ;  $\kappa \sim \text{Beta}$ ;  $\sigma_L \sim \text{HalfNormal}$  (following the recommendations of Gelman 2006). Hyperparameters are calibrated against the AmP collection ranges (Marques *et al.* 2018) and passed to Stan as data, so that changing priors does not require recompilation.

**Prior calibration rationale.** Table 1 shows the *generic* default hyperparameters (`prior_default()`) and their implied 95% prior mass ranges. These are deliberately wide ( $\sigma = 1.0$ ) to accommodate any species; they are *not* suitable for quantitative inference without tightening (see Section 6.4).

For common test species, the function `prior_species()` returns priors calibrated to the AmP collection (Marques *et al.* 2018), with  $\sigma = 0.3$  for well-characterised parameters:

```
R> prior_species("Eisenia_fetida")
R> prior_species("Daphnia_magna", type = "debtox")
```

Built-in species include *Eisenia fetida*, *E. andrei*, *Folsomia candida*, *Daphnia magna*, and *Lumbricus rubellus*. Users working with other species should construct custom priors from the relevant AmP entry. We recommend always conducting a prior sensitivity analysis (Section 6.4).

### 3.4. Joint posterior

The posterior is:

$$p(\boldsymbol{\theta} \mid \mathbf{y}) \propto p(\boldsymbol{\theta}) \prod_{i=1}^{N_L} p(L_i^{\text{obs}} \mid \hat{L}_i(\boldsymbol{\theta}), \sigma_L) \prod_{k=1}^{N_R} p(R_k \mid \Delta E_{R,k}(\boldsymbol{\theta}), k_R, \phi), \quad (14)$$

sampled via the No-U-Turn Sampler (NUTS; Hoffman and Gelman 2014) with a stiff BDF ODE solver evaluated at each leapfrog step.

Table 1: Default prior specifications and implied ranges, compared with AmP empirical ranges for earthworms and springtails.

| Parameter    | Prior family | $\mu / a$ | $\sigma / b$ | 95% prior range |
|--------------|--------------|-----------|--------------|-----------------|
| $\{p_{Am}\}$ | LogNormal    | 1.0       | 1.0          | 0.4–20          |
| $[p_M]$      | LogNormal    | –1.0      | 1.0          | 0.04–2          |
| $\kappa$     | Beta         | 2         | 2            | 0.09–0.91       |
| $v$          | LogNormal    | –1.5      | 1.0          | 0.02–1          |
| $[E_G]$      | LogNormal    | 6.0       | 1.0          | 100–3000        |
| $\sigma_L$   | HalfNormal   | 0         | 0.1          | 0–0.2           |

### 3.5. Hierarchical extension

For  $J$  individuals with individual random effects on  $\{p_{Am}\}$ , the hierarchical model adds population-level hyperparameters  $\mu_{\log\{p_{Am}\}}$  and  $\sigma_{\log\{p_{Am}\}}$ :

$$\log\{p_{Am}\}_j \mid \mu, \sigma \sim \mathcal{N}(\mu, \sigma), \quad j = 1, \dots, J. \quad (15)$$

The non-centred reparameterisation  $\log\{p_{Am}\}_j = \mu + \sigma \cdot z_j$ ,  $z_j \sim \mathcal{N}(0, 1)$  avoids the pathological funnel geometry (Betancourt and Girolami 2015).

## 4. BayesianDEB: Software design

### 4.1. Architecture

The package consists of three layers:

**Stan models.** Four pre-written `.stan` programs in `inst/stan/`, implementing the DEB ODE system with the variadic `ode_bdf_tol` solver (BDF, tolerances  $10^{-6}$ ). The hierarchical and DEBtox models use `reduce_sum` for within-chain parallelism. Observation model switching (Gaussian / log-normal / Student- $t$  for growth; negative binomial / Poisson for reproduction) is controlled by integer flags in the **Stan** data block, avoiding recompilation when changing observation families or prior hyperparameters. Structural model changes (e.g., adding state variables or reparametrising in terms of derived quantities) require modifying the underlying **Stan** programs.

**R interface.** Exported functions for data preparation (`bdeb_data`), model specification (`bdeb_model`), MCMC fitting (`bdeb_fit`), diagnostics (`bdeb_diagnose`), posterior predictive checks (`bdeb_ppc`), derived quantities (`bdeb_derived`), LOO cross-validation (`bdeb_loo`), prior predictive simulation (`bdeb_prior_predictive`), reproducibility reporting (`bdeb_session_info`), and standalone ODE simulators (`deb_simulate`, `debtox_simulate`). All functions use S3 classes (`bdeb_data`, `bdeb_model`, `bdeb_fit`) with `print`, `plot`, `summary`, `coef`, and `predict` methods.

**Prior system.** A family of `prior_*`() constructors (`prior_lognormal`, `prior_beta`, `prior_halfnormal`, `prior_halfcauchy`, `prior_exponential`, `prior_normal`) that encode distribution choices as lightweight S3 objects. Weakly informative defaults calibrated against the AmP collection (Marques *et al.* 2018) are provided via `prior_default()`.



## 4.2. Workflow

The recommended workflow follows the iterative Bayesian modelling cycle of [Gelman \*et al.\* \(2013, Ch. 6\)](#):

1. Prepare and validate data with `bdeb_data()`.
2. Specify model, priors, and observation family with `bdeb_model()`.
3. Fit via MCMC with `bdeb_fit()`.
4. Check convergence with `bdeb_diagnose()` ( $\hat{R}$ , ESS, divergences; [Vehtari \*et al.\* 2021](#)).
5. Perform posterior predictive checks with `bdeb_ppc()`.
6. Compute derived quantities with `bdeb_derived()`.
7. Compare models via `bdeb_loo()` ([Vehtari \*et al.\* 2017](#)).
8. Iterate: revise priors or model structure as needed.

## 4.3. Observation models

The observation model is selected via constructor functions passed to the `observation` argument of `bdeb_model()`:

```
R> mod <- bdeb_model(dat, type = "individual",
+   observation = list(growth = obs_student_t(nu = 5)))
```

The choice is encoded as an integer flag in the Stan data block (`obs_growth = 1/2/3`, `obs_repro = 1/2`) and dispatched via conditional blocks in the model and generated quantities sections. This avoids recompilation when switching families.

## 4.4. Temperature correction

If a `temperature` list is supplied to `bdeb_model()`, the Arrhenius correction factor  $c_T$  is computed in the Stan `transformed data` block and applied to all rate parameters inside the ODE. The posterior gives estimates at  $T_{\text{ref}}$ , making results comparable across experiments at different temperatures.

## 4.5. Within-chain parallelism

The hierarchical and DEBtox models distribute per-individual (or per-group) ODE solves across threads via Stan's `reduce_sum`. The `threads_per_chain` argument in `bdeb_fit()` controls this:

```
R> fit <- bdeb_fit(mod, chains = 4, threads_per_chain = 4)
```

# 5. Illustrations



We illustrate three of the four model types: individual growth, hierarchical multi-individual growth, and DEBtox. The "growth\_repro" model follows the same interface with an additional reproduction data argument; its usage is documented in the package vignette and follows the same workflow as the individual model with the addition of offspring count data and the `obs_negbinomial()` or `obs_poisson()` observation family.

### 5.1. Individual growth: *Eisenia fetida*

The bundled `eisenia_growth` dataset contains simulated weekly length measurements for 21 individuals of the earthworm *Eisenia fetida* over 84 days, using AmP parameters ( $\{p_{Am}\} = 5.0$ ,  $[p_M] = 0.5$ ,  $\kappa = 0.75$ ,  $v = 0.2$ ,  $[E_G] = 400$ ) with individual variation in  $\{p_{Am}\}$  (CV  $\approx 10\%$ ) and Gaussian measurement error ( $\sigma_L = 0.015$  cm).

```
R> library("BayesianDEB")
R> data("eisenia_growth")
R> df1 <- eisenia_growth[eisenia_growth$id == 5, ]
R> dat <- bdeb_data(growth = df1, f_food = 1.0)
R> mod <- bdeb_model(dat, type = "individual",
+   priors = list(
+     p_Am = prior_lognormal(mu = 1.5, sigma = 0.5),
+     kappa = prior_beta(a = 3, b = 2),
+     sigma_L = prior_halfnormal(sigma = 0.05)))
R> fit <- bdeb_fit(mod, chains = 4, iter_sampling = 2000,
+   adapt_delta = 0.9, seed = 42)
R> bdeb_diagnose(fit)
R> plot(fit, type = "trajectory", n_draws = 200, seed = 1)
```

The derived ultimate length and von Bertalanffy growth rate are obtained with full posterior uncertainty propagation:

```
R> bdeb_derived(fit, quantities = c("L_m", "L_inf", "k_M",
+   "g", "growth_rate"), f = 1.0)
```

All formulas follow [Kooijman \(2010, Table 3.1\)](#):  $L_m = \kappa\{p_{Am}\}/[p_M]$ ,  $L_\infty = f \cdot L_m$ ,  $k_M = [p_M]/[E_G]$ ,  $g = [E_G]v/(\kappa\{p_{Am}\})$ ,  $\dot{r}_B = k_M g/(3(f + g))$ .

**Model comparison via LOO.** The observation model can be compared via approximate leave-one-out cross-validation ([Vehtari et al. 2017](#)):

```
R> mod_ln <- bdeb_model(dat, type = "individual",
+   observation = list(growth = obs_lognormal()),
+   priors = list(
+     p_Am = prior_lognormal(mu = 1.5, sigma = 0.5),
+     kappa = prior_beta(a = 3, b = 2),
+     sigma_L = prior_halfnormal(sigma = 0.05)))
R> fit_ln <- bdeb_fit(mod_ln, chains = 4,
+   iter_sampling = 2000, adapt_delta = 0.9, seed = 42)
```

```
R> loo_gauss <- bdeb_loo(fit)
R> loo_ln <- bdeb_loo(fit_ln)
R> loo::loo_compare(list(gaussian = loo_gauss,
+   lognormal = loo_ln))
```

## 5.2. Hierarchical model

```
R> dat_all <- bdeb_data(growth = eisenia_growth, f_food = 1.0)
R> mod_h <- bdeb_model(dat_all, type = "hierarchical",
+   priors = list(
+     mu_log_p_Am = prior_normal(mu = 1.5, sigma = 0.5),
+     sigma_log_p_Am = prior_exponential(rate = 2)))
R> fit_h <- bdeb_fit(mod_h, chains = 4, adapt_delta = 0.95,
+   threads_per_chain = 4, seed = 123)
R> bdeb_diagnose(fit_h)
R> plot(fit_h, type = "trajectory", n_draws = 50, seed = 1)
```

The trajectory plot produces a faceted display with one panel per individual, showing posterior predicted curves overlaid on observed data. The hierarchical model yields narrower credible intervals for shared parameters than the individual fit because it pools information across 21 organisms.

## 5.3. DEBtox analysis

```
R> data("debtox_growth")
R> conc_map <- setNames(unique(debtox_growth$concentration),
+   as.character(unique(debtox_growth$concentration)))
R> dat_tox <- bdeb_data(growth = debtox_growth,
+   concentration = conc_map, f_food = 1.0)
R> mod_tox <- bdeb_tox(dat_tox, stress = "assimilation",
+   priors = list(
+     z_w = prior_lognormal(mu = 2.5, sigma = 1.0),
+     b_w = prior_lognormal(mu = -5.0, sigma = 2.0)))
R> fit_tox <- bdeb_fit(mod_tox, chains = 4, adapt_delta = 0.95,
+   threads_per_chain = 2, seed = 77)
R> bdeb_ec50(fit_tox, prob = 0.95)
R> plot_dose_response(fit_tox, n_draws = 100, seed = 1)
```

The  $EC_{50}$  is computed analytically in Stan's `generated quantities` block as  $z_w + 0.5/b_w$ , giving the full posterior distribution without post-hoc root-finding.

**Real-data DEBtox: cadmium toxicity to *E. andrei*.** To demonstrate the DEBtox model on experimental data, we analyse the growth of *Eisenia andrei* exposed to cadmium (0, 10, 32, 100, 320 mg Cd kg<sup>-1</sup> in natural soil) from [Gestel \*et al.\* \(1991\)](#), obtained via the EGGrowth database ([Mathieu 2018](#)). The dataset comprises 5 concentration groups with 7 time

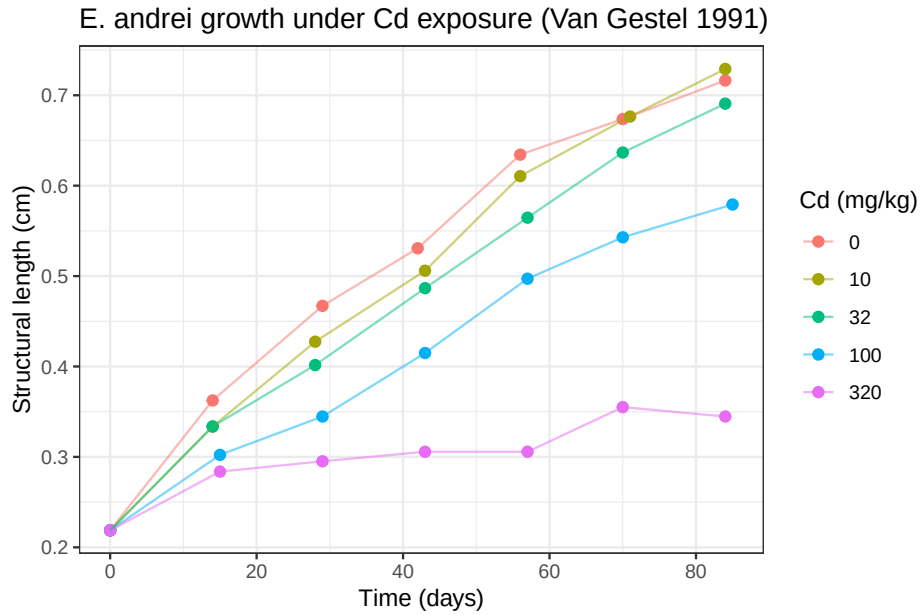


Figure 1: Growth of *E. andrei* under cadmium exposure (Gestel *et al.* 1991): structural length over 84 days at 5 concentrations. Data from EGrowth (Mathieu 2018).

points each over 84 days (group-mean body mass converted to structural length and bundled as `eisenia_cd`).

```
R> data("eisenia_cd")
R> conc_map <- setNames(unique(eisenia_cd$concentration),
+   as.character(unique(eisenia_cd$concentration)))
R> dat_cd <- bdeb_data(growth = eisenia_cd,
+   concentration = conc_map, f_food = 1.0)
R> mod_cd <- bdeb_tox(dat_cd, stress = "assimilation",
+   priors = list(
+     z_w = prior_lognormal(mu = 3.0, sigma = 1.0),
+     b_w = prior_lognormal(mu = -4.0, sigma = 1.5)))
R> fit_cd <- bdeb_fit(mod_cd, chains = 4,
+   adapt_delta = 0.95, seed = 42, init = 0.5)
R> bdeb_ec50(fit_cd, prob = 0.90)
```

The model converged without divergences ( $\hat{R} < 1.02$  for all parameters). The posterior  $EC_{50}$  for growth inhibition is  $225 \text{ mg Cd kg}^{-1}$  (90% CI: [181, 257]), consistent with published 84-day  $EC_{50}$  values of  $100\text{--}300 \text{ mg Cd kg}^{-1}$  for lumbricid earthworms in natural soils (Gestel *et al.* 1991). The NEC (no-effect concentration) is estimated at  $6.2 \text{ mg Cd kg}^{-1}$  ([1.8, 15.6]), below the lowest tested concentration ( $10 \text{ mg kg}^{-1}$ ), indicating that even the lowest treatment may have induced marginal stress. Table 2 reports the full posterior summary. This analysis demonstrates that **BayesianDEB** can recover ecotoxicologically meaningful effect thresholds with full posterior uncertainty from real experimental data. Figure 2 shows the dose-response relationship.

Table 2: DEBtox posterior summary for cadmium toxicity to *E. andrei* (Gestel *et al.* 1991). 4 chains, 2000 iterations, 0 divergences.

| Parameter        | Median | 90% CI         | $\hat{R}$ | Bulk ESS |
|------------------|--------|----------------|-----------|----------|
| $\{p_{Am}\}$     | 1.59   | [0.61, 4.84]   | 1.01      | 825      |
| $[p_M]$          | 0.43   | [0.11, 1.17]   | 1.00      | 892      |
| $\kappa$         | 0.61   | [0.27, 0.90]   | 1.00      | 3855     |
| $k_d$            | 0.38   | [0.09, 1.81]   | 1.01      | 592      |
| $z_w$ (NEC)      | 6.15   | [1.82, 15.6]   | 1.00      | 5465     |
| $b_w$            | 0.0023 | [0.002, 0.003] | 1.01      | 480      |
| $\sigma_L$       | 0.024  | [0.019, 0.031] | 1.01      | 1546     |
| EC <sub>50</sub> | 225    | [181, 257]     | 1.01      | 502      |

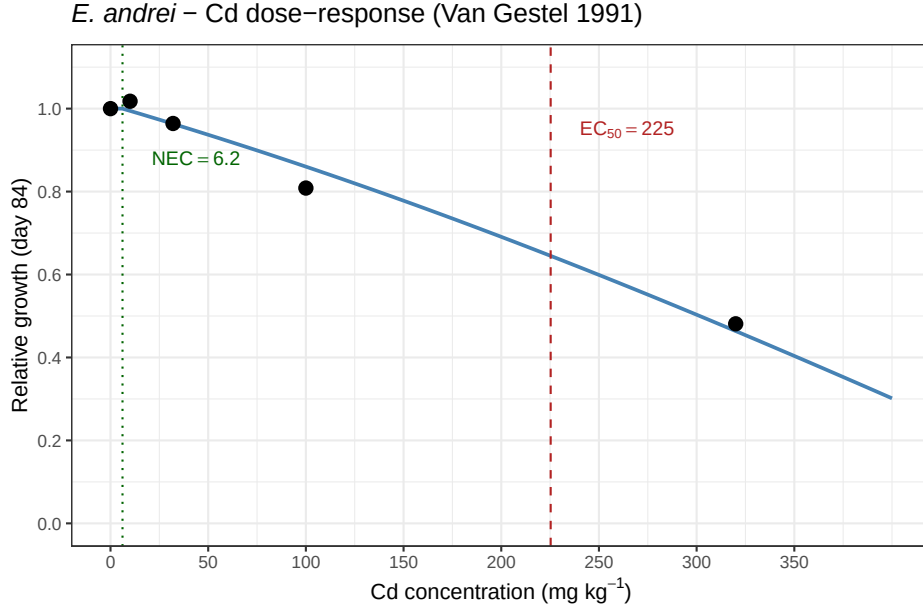


Figure 2: Dose-response curve for *E. andrei* growth under cadmium exposure (Gestel *et al.* 1991). Points: observed relative growth at day 84 (normalised to control). Blue line: DEBtox model prediction at posterior medians. Dashed red: EC<sub>50</sub> = 225 mg Cd kg<sup>-1</sup>; dotted green: NEC = 6.2 mg Cd kg<sup>-1</sup>.

## 6. Validation and performance

### 6.1. Simulation-based calibration

We validate the inference pipeline with simulation-based calibration (SBC; Talts *et al.* 2018). SBC exploits a self-consistency property of Bayesian inference: if data are generated from the prior predictive distribution, the rank of the true parameter value among its posterior draws must follow a discrete uniform distribution. Non-uniform rank histograms diagnose specific failures — humps indicate over-concentration, U-shapes over-dispersion, and asymmetry in-

Table 3: Priors used for SBC, calibrated to *E. fetida* AmP parameter ranges.

| Parameter    | Family     | $\mu / a$ | $\sigma / b$ |
|--------------|------------|-----------|--------------|
| $\{p_{Am}\}$ | LogNormal  | 1.5       | 0.5          |
| $[p_M]$      | LogNormal  | -1.0      | 0.5          |
| $\kappa$     | Beta       | 3         | 2            |
| $v$          | LogNormal  | -1.5      | 0.5          |
| $[E_G]$      | LogNormal  | 6.0       | 0.5          |
| $E_0$        | LogNormal  | 0.0       | 0.5          |
| $L_0$        | LogNormal  | -2.0      | 0.5          |
| $\sigma_L$   | HalfNormal | 0         | 0.05         |

icates bias. Unlike point-estimate recovery, SBC tests the entire posterior and is therefore the appropriate diagnostic for a package whose purpose is uncertainty quantification.

**Protocol.** For each replication  $r = 1, \dots, 500$ :

1. Draw  $\theta^{*(r)}$  from the prior  $\pi(\theta)$  (Table 3).
2. Forward-simulate a DEB growth trajectory via the LSODA solver (**deSolve**, tolerances  $10^{-6}$ ) and generate  $N_t = 13$  weekly length observations with Gaussian noise  $L_i^{\text{obs}} \sim \mathcal{N}(\hat{L}_i, \sigma_L^{*(r)})$ .
3. Fit the "individual" model with `bdeb_fit()` (4 chains, 500 warmup + 500 sampling,  $\delta = 0.9$ ); thin pooled draws to  $L = 199$ .
4. Discard the replication if any divergent transitions occurred or  $\hat{R} > 1.05$  for any parameter.
5. Compute the rank  $\rho_j^{(r)} = \#\{l : \theta_j^{(l)} < \theta_j^{*(r)}\}$  for each parameter  $j$ .

The LSODA solver in **deSolve** automatically switches between stiff (BDF) and non-stiff (Adams) methods; for the mildly stiff DEB system, it predominantly uses BDF steps that are methodologically equivalent to Stan's `ode_bdf_tol1`. Both solvers use matching tolerances ( $10^{-6}$ ). We verified numerical consistency by comparing trajectories from both solvers for 100 random parameter draws: the maximum absolute deviation in predicted length was below  $10^{-5}$  cm in all cases.

**Results.** Of the 500 replications, 397 converged without divergences (64 were skipped due to degenerate prior draws that produced biologically implausible trajectories, 39 failed convergence checks). Figure 3 shows the rank histograms for six core DEB parameters; all panels are visually consistent with uniformity (dashed reference line). Table 4 reports  $\chi^2$  goodness-of-fit tests (20 bins, 19 df) and empirical credible interval coverage. No parameter rejects uniformity at  $\alpha = 0.05$  (minimum  $p = 0.31$  for  $[p_M]$ ), and empirical 90% coverage ranges from 0.87 to 0.91, within simulation error of the nominal level. This confirms that, conditional on successful MCMC convergence, the **BayesianDEB** individual growth model is well-calibrated. We note that 21% of replications (103/500) were discarded; the calibration

Table 4: SBC results:  $\chi^2$  uniformity test (20 bins, 19 df) and empirical credible interval coverage (397 valid replications out of 500,  $L = 199$ ). No parameter rejects uniformity at  $\alpha = 0.05$ .

| Parameter    | $\chi^2_{19}$ | $p$ value | Cov. 50% | Cov. 90% | Cov. 95% |
|--------------|---------------|-----------|----------|----------|----------|
| $\{p_{Am}\}$ | 12.1          | 0.88      | 0.50     | 0.89     | 0.92     |
| $[p_M]$      | 21.5          | 0.31      | 0.53     | 0.91     | 0.94     |
| $\kappa$     | 18.0          | 0.52      | 0.54     | 0.90     | 0.94     |
| $v$          | 16.7          | 0.61      | 0.48     | 0.87     | 0.92     |
| $[E_G]$      | 14.3          | 0.76      | 0.48     | 0.87     | 0.93     |
| $\sigma_L$   | 20.1          | 0.39      | 0.47     | 0.88     | 0.93     |

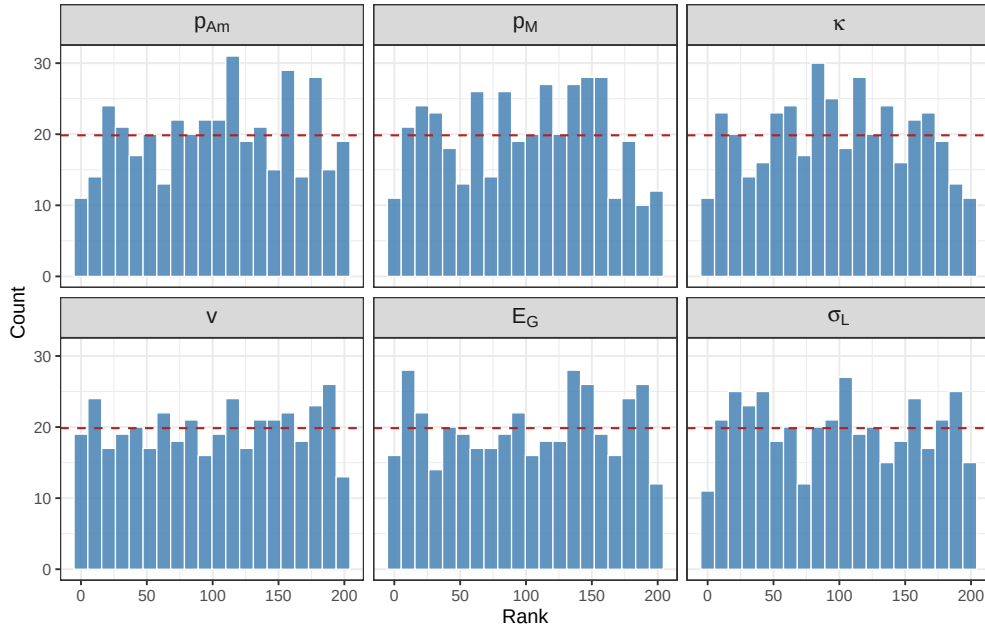


Figure 3: SBC rank histograms for six DEB parameters (397 valid replications out of 500 attempts, 20 bins). Dashed line: expected count under uniformity. Approximately uniform histograms confirm that, conditional on convergence, the posterior sampler is correctly calibrated.

result therefore applies to the subset of parameter space where MCMC converges successfully, not unconditionally to all possible inputs.

The full SBC script (`sbc_validation.R`) and archived results (`sbc_results.rds`) are provided in the replication materials.

**Hierarchical model SBC.** We repeated the SBC procedure for the hierarchical model with 5 individuals per replicate (100 attempts, 38 valid; Table 5). The low yield (38%) reflects the additional complexity of per-individual ODE solves under extreme prior draws. Given the small effective sample, the  $\chi^2$  tests have limited statistical power (expected count  $\approx 3.8$  per bin with 10 bins); these results should therefore be interpreted as *preliminary evidence*

Table 5: SBC results for the hierarchical model (38 valid replications out of 100 attempts, 10 bins).

| Parameter                 | $\chi^2$ | $p$ value | Cov. 50% | Cov. 90% | Cov. 95% |
|---------------------------|----------|-----------|----------|----------|----------|
| $\mu_{\log\{p_{Am}\}}$    | 3.6      | 0.94      | 0.61     | 0.95     | 0.95     |
| $\sigma_{\log\{p_{Am}\}}$ | 10.4     | 0.32      | 0.55     | 0.87     | 0.95     |
| $[p_M]$                   | 9.9      | 0.36      | 0.55     | 0.92     | 0.92     |
| $\kappa$                  | 5.7      | 0.77      | 0.50     | 0.87     | 0.95     |
| $\sigma_L$                | 16.7     | 0.05      | 0.74     | 1.00     | 1.00     |

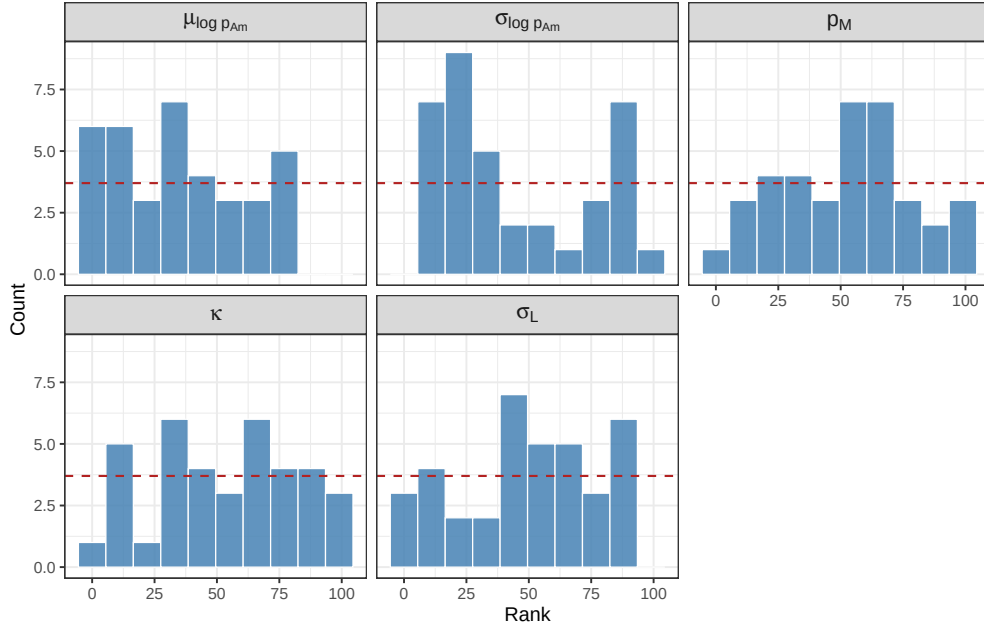


Figure 4: SBC rank histograms for the hierarchical model (38 valid replications, 10 bins). The dashed line indicates the expected count under uniformity.

of calibration, not as definitive validation. No parameter rejects uniformity at  $\alpha = 0.05$ , though  $\sigma_L$  is borderline ( $p = 0.053$ ) with overcoverage at the 90% and 95% levels (Table 5), suggesting possible conservatism in the  $\sigma_L$  posterior for small datasets. A more comprehensive hierarchical SBC with  $\geq 200$  attempts is planned for a future version.

## 6.2. Posterior correlation structure and identifiability

DEB models exhibit strong posterior correlations arising from the  $\kappa$ -rule structure. We characterise these quantitatively using the posterior correlation matrix from the individual model fit.

The dominant correlations are between  $[p_M]$  and  $v$  ( $r = 0.85$ ) and between  $[p_M]$  and  $[E_G]$  ( $r = 0.80$ ), reflecting the fact that growth dynamics jointly constrain maintenance, energy conductance, and structural costs but cannot resolve the individual contributions of these rate parameters. The strong  $\{p_{Am}\}$ – $[E_G]$  correlation ( $r = 0.73$ ) arises because both parameters



Table 6: Posterior rank correlation matrix (Spearman) for the individual growth model. Correlations  $|r| < 0.2$  are suppressed for clarity.

|              | $\{p_{Am}\}$ | $[p_M]$ | $\kappa$ | $v$   | $[E_G]$ | $\sigma_L$ |
|--------------|--------------|---------|----------|-------|---------|------------|
| $\{p_{Am}\}$ | 1.00         | 0.31    |          | 0.26  | 0.73    |            |
| $[p_M]$      | 0.31         | 1.00    |          | 0.85  | 0.80    |            |
| $\kappa$     |              |         | 1.00     | -0.35 | 0.21    |            |
| $v$          | 0.26         | 0.85    | -0.35    | 1.00  | 0.63    |            |
| $[E_G]$      | 0.73         | 0.80    | 0.21     | 0.63  | 1.00    |            |
| $\sigma_L$   |              |         |          |       |         | 1.00       |

control the rate of structural investment: higher assimilation can compensate for higher costs of structure, producing near-identical growth trajectories. The correlation between  $\kappa$  and  $v$  ( $r = -0.35$ ) reflects the  $\kappa$ -rule trade-off between somatic allocation and reserve mobilisation. By contrast, the  $\{p_{Am}\}$ – $[p_M]$  correlation is only moderate ( $r = 0.31$ ), indicating that these two parameters are less directly confounded than the theoretical relationship  $L_\infty = \kappa\{p_{Am}\}/[p_M]$  might suggest — the data constrain their ratio primarily through the indirect action of  $[E_G]$  and  $v$ .

*Practical consequences.* (i) Marginal credible intervals for individual rate parameters ( $\{p_{Am}\}$ ,  $[p_M]$ ,  $v$ ) are wide and substantially prior-driven. (ii) Derived quantities computed via `bdeb_derived()` propagate correlations correctly and are *somewhat* better identified in relative terms, but they are not immune to prior sensitivity — particularly  $L_\infty$  and  $\dot{r}_B$ , which show near-zero contraction under moderately informative priors (Table 7) and shift substantially between prior specifications (Table 9). (iii) Informative priors on  $\kappa$  and  $[E_G]$  substantially improve identifiability. We recommend always inspecting `plot(fit, type = "pairs")` and reporting both derived quantities *and* their prior sensitivity.

**BayesianDEB** should not be used for quantitative parameter inference from growth-only data with a single individual unless species-specific informative priors or hierarchical data structures are employed. Without these, posteriors for rate parameters will be substantially prior-driven.

See Figure 8 for the bivariate posterior structure.

**Prior-to-posterior contraction.** To quantify how much the data inform each parameter beyond the prior, we define the *contraction ratio*

$$c_j = 1 - \frac{w_j^{\text{post}}}{w_j^{\text{prior}}}, \quad (16)$$

where  $w_j$  denotes the 90% credible interval width for parameter  $\theta_j$ . Values near 0 indicate a prior-dominated posterior; values near 1 indicate data-dominated inference. Table 7 reports  $c_j$  for the individual growth model fitted to simulated *E. fetida* data (84 days,  $f = 1$ ) with varying numbers of observations, using the moderately informative priors of Table 3.

Table 7 reveals that for individual-level growth data, only  $\sigma_L$  is well-determined by the data ( $c > 0.8$ ); all DEB rate parameters remain largely prior-dominated ( $c < 0.15$ ) even with 37 time points.  $[E_G]$  shows moderate contraction ( $c \approx 0.35$ ) under informative priors, as it

Table 7: Prior-to-posterior contraction  $c$  for different data sizes and prior widths (individual growth model, simulated *E. fetida* data,  $N_t = 13$ ). Columns A use moderately informative priors ( $\sigma = 0.5$ ); column B uses the package defaults ( $\sigma = 1.0$ ). Values near 0 indicate prior-dominated posteriors; near 1 data-dominated. The entry for  $[E_G]$  under default priors is omitted because the default  $[E_G]$  prior ( $\sigma = 1.0$ ) produced frequent ODE solver failures during MCMC warmup, preventing reliable estimation.

| Parameter    | Informative ( $\sigma = 0.5$ ) |            |            | Default ( $\sigma = 1.0$ ) |                    |
|--------------|--------------------------------|------------|------------|----------------------------|--------------------|
|              | $N_t = 5$                      | $N_t = 13$ | $N_t = 37$ | $N_t = 13$                 | Status             |
| $\{p_{Am}\}$ | 0.00                           | 0.00       | 0.08       | 0.00                       | prior-dominated    |
| $[p_M]$      | 0.00                           | 0.03       | 0.03       | 0.17                       | prior-dominated    |
| $\kappa$     | 0.11                           | 0.11       | 0.14       | 0.07                       | weakly informed    |
| $v$          | 0.09                           | 0.10       | 0.10       | 0.00                       | prior-dominated    |
| $[E_G]$      | 0.33                           | 0.35       | 0.38       | —                          | partially informed |
| $\sigma_L$   | 0.67                           | 0.87       | 0.94       | 0.93                       | data-dominated     |
| $L_\infty$   | 0.00                           | 0.00       | 0.07       | 0.00                       | prior-dominated    |
| $\dot{r}_B$  | 0.02                           | 0.08       | 0.14       | 0.00                       | weakly informed    |

jointly controls growth curvature with  $v$ . The derived quantities  $L_\infty$  and  $\dot{r}_B$  show similarly low contraction because the moderately informative priors already constrain the feasible parameter combinations. Increasing  $N_t$  from 5 to 37 primarily improves  $\sigma_L$ ; the rate parameters show negligible benefit (Figure 5).

The rightmost column of Table 7 shows the contraction under the package *default* priors ( $\sigma = 1.0$ ). The pattern is even starker:  $\{p_{Am}\}$ ,  $v$ ,  $L_\infty$ , and  $\dot{r}_B$  all show zero contraction, meaning the posterior is indistinguishable from the prior. Only  $\sigma_L$  remains data-dominated ( $c = 0.93$ ). This underscores the importance of using species-specific priors (`prior_species()`) rather than defaults for any quantitative inference.

*This means that a user applying **BayesianDEB** with weakly informative priors to single-individual growth data should expect posteriors that are substantially shaped by the prior, not the data.* This is not a deficiency of the software but a fundamental identifiability limitation of growth-only DEB models: growth curves constrain the observation error and, partially, the energy costs, but cannot disentangle the individual contributions of  $\{p_{Am}\}$ ,  $[p_M]$ ,  $v$ , and  $\kappa$  to the observed trajectory. Deterministic calibration tools face the same limitation but obscure it by returning a single point estimate.

Three strategies mitigate this:

1. *Species-specific priors.* Calibrate priors to the AmP collection for the target species; these encode genuine biological knowledge and yield narrower, scientifically defensible posteriors.
2. *Hierarchical modelling.* The "`hierarchical`" model pools information across individuals, substantially improving identifiability of the shared parameters  $[p_M]$ ,  $\kappa$ ,  $v$ ,  $[E_G]$ .
3. *Prior sensitivity analysis.* Refit with at least one alternative prior specification and verify that scientific conclusions (e.g.,  $L_\infty$ ,  $EC_{50}$ ) are robust.

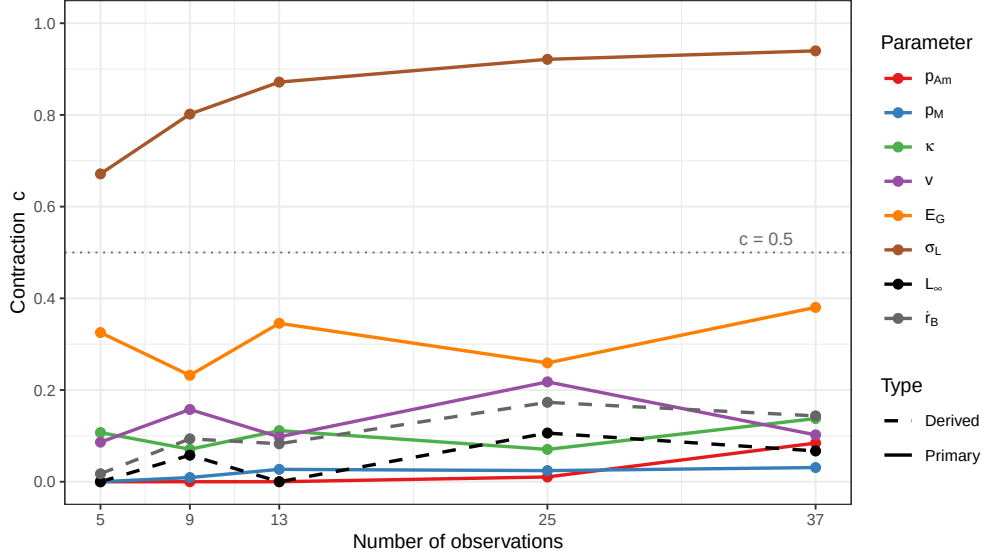


Figure 5: Prior-to-posterior contraction as a function of the number of observations. Solid: primary parameters; dashed: derived quantities. Only  $\sigma_L$  (and partially  $[E_G]$ ) are data-determined; all rate parameters remain prior-dominated even with dense sampling. The dotted line marks  $c = 0.5$ .

We recommend reporting the contraction ratio  $c$  alongside posterior summaries as a routine diagnostic. A contraction  $c < 0.2$  signals a prior-dominated parameter that warrants careful interpretation.

### 6.3. Real data case study: *Eisenia fetida* growth

To demonstrate the package on real experimental data, we analyse the growth of *Eisenia fetida* on activated sludge at 25 °C from Neuhauser *et al.* (1980), obtained via the EGrowth database (Mathieu 2018). The dataset comprises 37 group-mean body mass measurements (20 worms per replicate) over 250 days, from hatchling ( $\sim 8$  mg) to adult ( $\sim 2350$  mg). Wet mass was converted to structural length via  $L = (W/d_V)^{1/3}$  with tissue density  $d_V = 1.05 \text{ g cm}^{-3}$  (assuming  $\delta_M = 1$ , i.e., no shape correction) and bundled as `eisenia_neuhauser`.

**Group-mean data.** The Neuhauser dataset reports group-mean body mass (20 worms per replicate), not individual-level observations. Fitting the "individual" model to group means treats each mean as a single noisy measurement of the underlying DEB trajectory. This is a common pragmatic choice when individual-level data are unavailable, but it has two consequences: (i) the observation error  $\sigma_L$  conflates measurement noise with among-individual variation, and (ii) the effective sample size for estimating  $\sigma_L$  equals the number of time points ( $N_t = 37$ ), not the number of organisms. For datasets where individual-level records are available, the "hierarchical" model is preferable as it explicitly separates within- and between-individual variation. A similar caveat applies to the DEBtox analysis (Section 5.3), which uses concentration-group means.

Table 8: Posterior estimates from the Neuhauser (1980) dataset vs. AmP reference values for *E. fetida*. Posterior at  $T_{\text{ref}} = 293.15$  K.

| Parameter    | Posterior median | 90% CI       | AmP value | Units                            |
|--------------|------------------|--------------|-----------|----------------------------------|
| $\{p_{Am}\}$ | 5.00             | [2.91, 9.08] | 3.9–6.2   | $\text{J d}^{-1} \text{cm}^{-2}$ |
| $[p_M]$      | 0.64             | [0.40, 1.02] | 0.3–0.8   | $\text{J d}^{-1} \text{cm}^{-3}$ |
| $\kappa$     | 0.82             | [0.58, 0.96] | 0.6–0.85  | —                                |
| $v$          | 1.50             | [0.91, 2.50] | 0.1–0.3   | $\text{cm d}^{-1}$               |
| $L_\infty$   | 6.86             | [3.52, 12.8] | 5–12      | cm                               |

```
R> library("BayesianDEB")
R> data("eisenia_neuhauser")
R> dat <- bdeb_data(growth = eisenia_neuhauser, f_food = 1.0)
R> mod <- bdeb_model(dat, type = "individual",
+   priors = list(
+     p_Am = prior_lognormal(mu = 1.5, sigma = 0.5),
+     kappa = prior_beta(a = 3, b = 2)),
+   temperature = list(T = 298.15, T_ref = 293.15, T_A = 8000))
R> fit <- bdeb_fit(mod, chains = 4, iter_sampling = 2000,
+   adapt_delta = 0.95, seed = 42)
R> bdeb_diagnose(fit)
```

The model converged without divergences ( $\hat{R} < 1.01$ , bulk ESS  $> 1000$  for all parameters). Table 8 compares the posterior estimates (at  $T_{\text{ref}} = 20^\circ\text{C}$ ) with AmP reference values for *E. fetida*.

**Discrepancy in  $v$ .** The posterior median for  $v$  ( $1.50 \text{ cm d}^{-1}$ ) exceeds the AmP reference range ( $0.1\text{--}0.3 \text{ cm d}^{-1}$ ) by an order of magnitude. This discrepancy arises from three compounding factors. (i) *Low identifiability*. The energy conductance is poorly identified from growth data alone ( $c \approx 0.10$ ; Table 7), and the default prior  $v \sim \text{LogNormal}(-1.5, 1.0)$  is wide enough to place non-negligible mass above  $1 \text{ cm d}^{-1}$  (prior 97.5th percentile  $\approx 1.6$ ). (ii) *Correlation with  $[p_M]$* . The posterior rank correlation between  $v$  and  $[p_M]$  is  $r = 0.85$  (Table 6): the  $\kappa$ -rule structure admits a compensating trade-off in which higher  $v$  and higher  $[p_M]$  produce near-identical growth trajectories. (iii) *Mass-to-length conversion*. The conversion  $L = (W/d_V)^{1/3}$  ignores reserve mass ( $W_E = d_E \cdot E$ ), overestimating structural volume and hence structural length. Because  $v$  scales the rate at which reserves are mobilised per unit length, an inflated length scale biases  $v$  upward.

This order-of-magnitude overestimate of  $v$  illustrates a key limitation: *individual marginal rate parameter estimates from growth-only data should not be interpreted as reliable point estimates, even when the model converges without divergences*. The  $v$  posterior is effectively unconstrained by the data ( $c \approx 0.10$ ; Table 7) and drifts to the tail of the prior under the influence of compensating correlations with  $[p_M]$ .

The derived quantities  $L_\infty = 6.86 \text{ cm}$  (AmP: 5–12) and  $\kappa = 0.82$  (AmP: 0.6–0.85) are broadly consistent with AmP, but as Section 6.4 shows, even  $L_\infty$  is sensitive to prior choice. This confirms the recommendation of Section 6.2: users should report derived quantities alongside

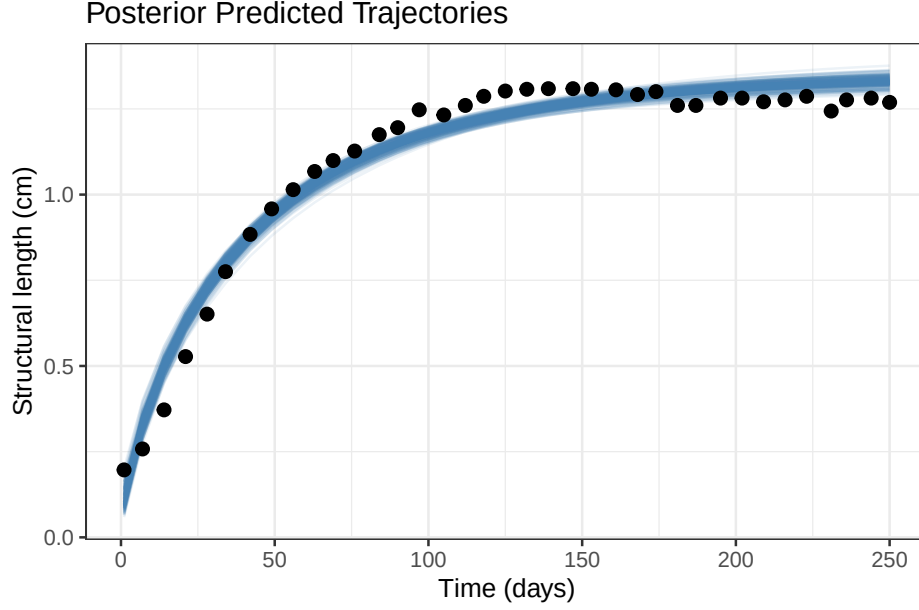


Figure 6: Posterior predicted growth trajectories (blue, 200 draws) overlaid on the observed group-mean data of Neuhauser *et al.* (1980) (black points). The DEB model captures the sigmoidal growth pattern.

their prior sensitivity, and employ species-specific priors (e.g.,  $v \sim \text{LogNormal}(\log 0.2, 0.3)$  calibrated to AmP) when individual parameter estimates are scientifically important.

```
R> plot(fit, type = "trajectory", n_draws = 200, seed = 1)
R> plot(bdeb_ppc(fit), n_draws = 200)
R> plot(fit, type = "pairs",
+       pars = c("p_Am", "p_M", "kappa", "E_G"))
```

Figure 6 shows the posterior predicted trajectories overlaid on the data, and Figure 7 the posterior predictive check.

The posterior pairs plot (Figure 8) reveals the expected  $\{p_{Am}\}-[E_G]$  and  $[p_M]-[E_G]$  correlations (Section 6.2).

#### 6.4. Prior sensitivity analysis

To demonstrate the effect of prior choice, we refit the bundled `eisenia_growth` data (individual 5, 13 weekly observations) under two specifications:

**Prior A (informative):**  $\{p_{Am}\}, [p_M], v, [E_G] \sim \text{LogNormal}(\mu, 0.5)$ ,  $\kappa \sim \text{Beta}(3, 2)$  — moderately informative, calibrated to the AmP collection.

**Prior B (weakly informative):** package defaults with  $\sigma = 1.0$  for all log-normal priors and  $\kappa \sim \text{Beta}(2, 2)$  — the specification a user obtains without manual prior calibration.

Table 9 compares the posterior summaries.

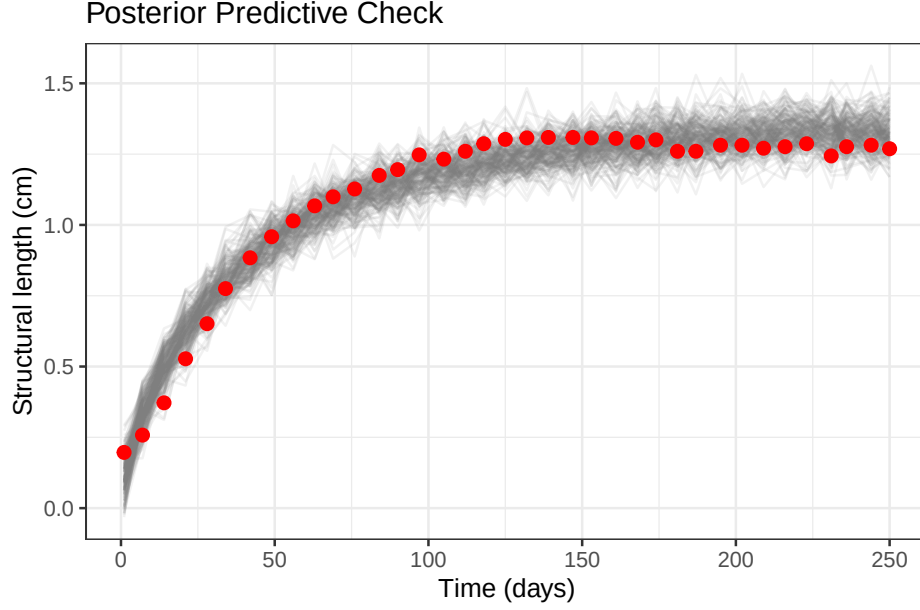


Figure 7: Posterior predictive check: replicated data from the fitted model (grey, 200 draws) vs. observed data (red). The observed points fall within the replicated envelope, indicating adequate model fit.

Table 9: Prior sensitivity: posterior medians and 90% CIs under informative (A,  $\sigma = 0.5$ ) and weakly informative (B,  $\sigma = 1.0$ ) priors. True values are known from the simulation.

| Parameter    | True  | Median A | 90% CI A         | Median B | 90% CI B         |
|--------------|-------|----------|------------------|----------|------------------|
| $\{p_{Am}\}$ | 5.0   | 4.42     | [1.96, 9.92]     | 3.02     | [0.66, 14.9]     |
| $[p_M]$      | 0.5   | 0.37     | [0.16, 0.85]     | 0.35     | [0.07, 1.63]     |
| $\kappa$     | 0.75  | 0.58     | [0.27, 0.88]     | 0.52     | [0.18, 0.86]     |
| $v$          | 0.2   | 0.20     | [0.10, 0.43]     | 0.24     | [0.06, 1.15]     |
| $\sigma_L$   | 0.015 | 0.012    | [0.009, 0.019]   | 0.013    | [0.009, 0.020]   |
| $L_\infty$   | 7.5   | 6.58     | [1.86, 22.6]     | 3.96     | [0.46, 42.3]     |
| $\dot{r}_B$  | —     | 0.0003   | [0.0001, 0.0008] | 0.0003   | [0.0000, 0.0023] |

Three patterns emerge. First,  $\sigma_L$  is insensitive to the prior (median  $\approx 0.012$ – $0.013$  in both fits); this is the only data-dominated parameter, consistent with the contraction analysis (Table 7). Second, marginal rate parameters shift modestly in median but their CIs widen by a factor of 2–3 under the weakly informative prior, confirming that these posteriors are substantially prior-shaped. Third, even  $L_\infty$  — a derived quantity expected to be better identified — changes from 6.58 to 3.96 and its 90% CI expands from [1.86, 22.6] to [0.46, 42.3]. This demonstrates that *with 13 growth observations and a single individual, no scientifically interesting quantity is immune to the prior.*

The practical conclusion is unambiguous: users must not rely on default priors for quantitative inference from individual-level growth data. Species-specific priors calibrated to the AmP collection or comparable literature are essential. We recommend always refitting with a

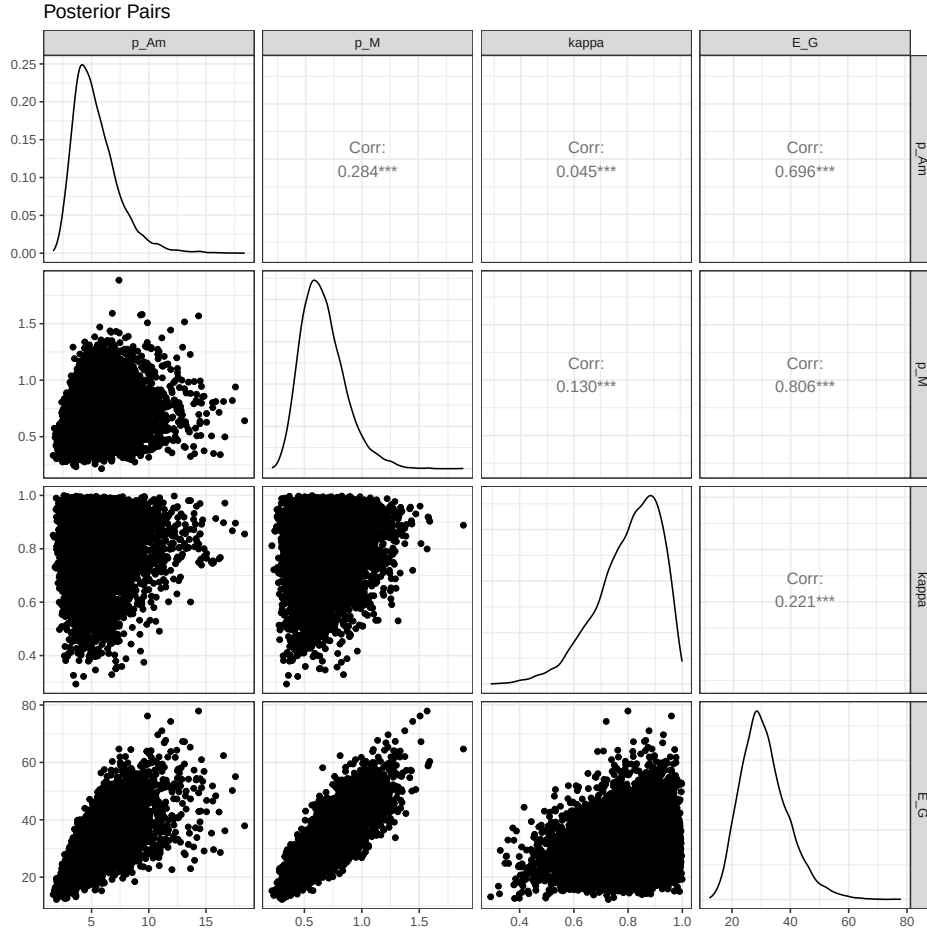


Figure 8: Posterior pairs plot showing bivariate correlations between  $\{p_{Am}\}$ ,  $[p_M]$ ,  $\kappa$ , and  $[E_G]$ . The strongest correlations in the full matrix are  $[p_M]-v$  ( $r = 0.85$ , not shown) and  $[p_M]-[E_G]$  ( $r = 0.80$ ); among the displayed parameters,  $\{p_{Am}\}-[E_G]$  ( $r = 0.73$ ) is the most prominent.

second prior specification and checking that the conclusions of interest (e.g.,  $L_\infty$  ranking across treatments,  $EC_{50}$  exceedance) are robust.

*Note:* the preceding sensitivity analysis used the simulated single-individual dataset (13 weekly observations). The diagnostic visualisations that follow (Section 6.5) return to the Neuhauser (1980) real-data fit (Section 6.3, 37 observations over 250 days).

## 6.5. Diagnostic visualisations

BayesianDEB produces several types of diagnostic plots, all returned as **ggplot2** objects for further customisation. Figure 9 shows MCMC trace plots from the Neuhauser (1980) *E. fetida* analysis, and Figure 10 shows the corresponding marginal posterior densities.

- **Trace plots** (type = "trace"): MCMC chains over iterations for visual convergence assessment. Well-mixed chains appear as overlapping “hairy caterpillars.”

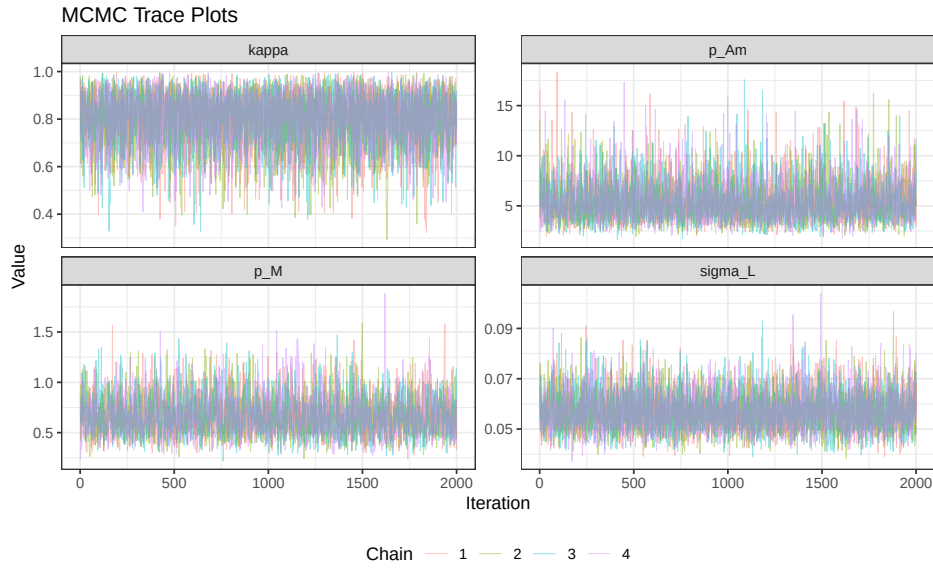


Figure 9: MCMC trace plots for core parameters ( $\{p_{Am}\}$ ,  $[p_M]$ ,  $\kappa$ ,  $\sigma_L$ ) from the Neuhauser (1980) fit. All four chains mix well ( $\hat{R} < 1.01$ ).

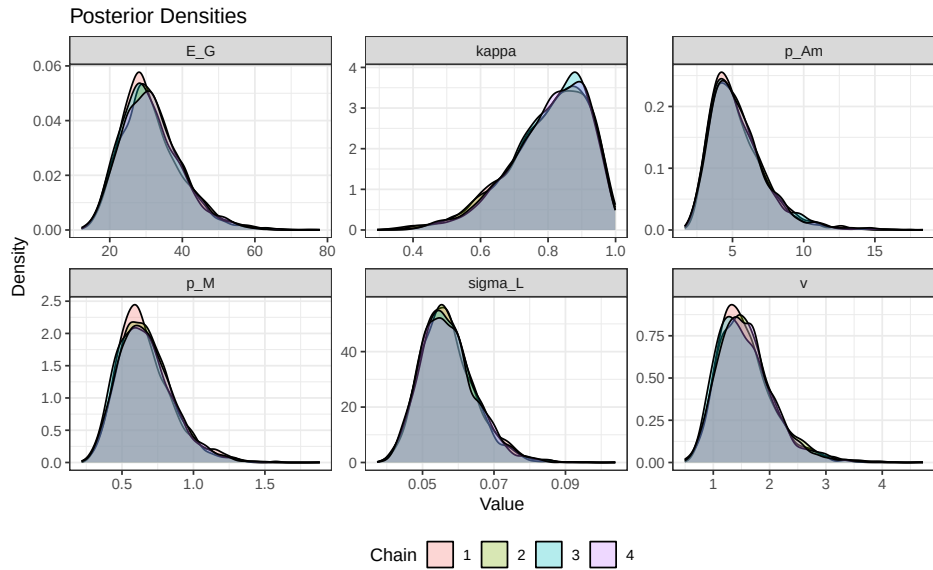


Figure 10: Marginal posterior densities for all six DEB parameters, coloured by chain. The overlap between chains confirms convergence.

- **Posterior densities** (`type = "posterior"`): Marginal density curves overlaid by chain, highlighting multi-modality or chain disagreement.
- **Pairs plots** (`type = "pairs"`): Bivariate posterior scatter plots revealing correlations. The  $[p_M]$ - $v$  and  $[p_M]$ - $[E_G]$  correlations are typically the strongest (Section 6.2).
- **Trajectory plots** (`type = "trajectory"`): Posterior predicted growth curves (blue)



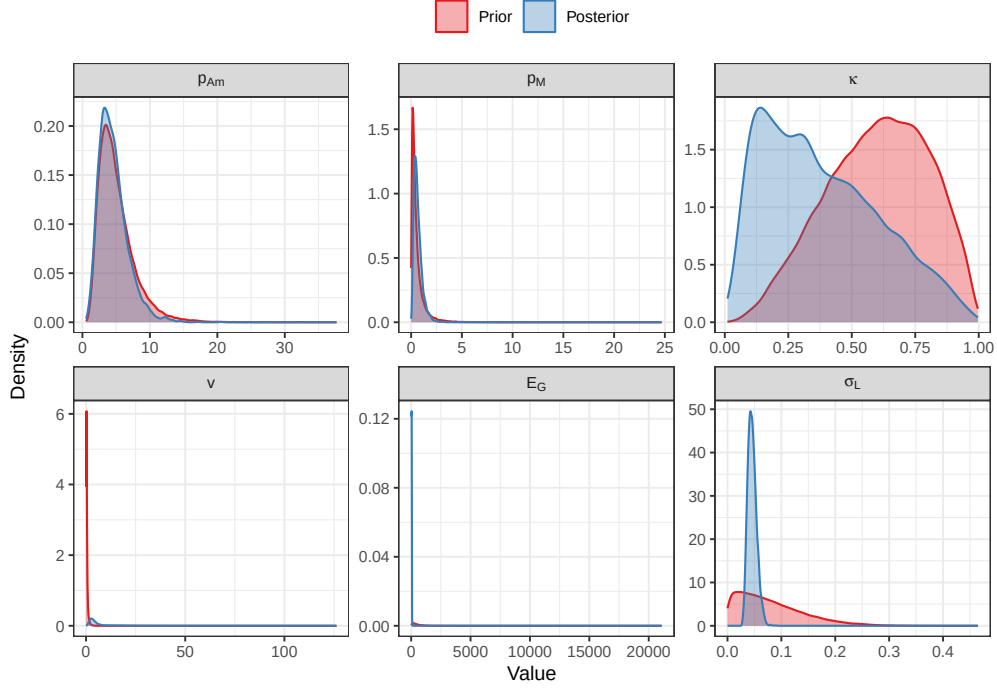


Figure 11: Prior (red) vs. posterior (blue) densities for six DEB parameters from the Neuhauser (1980) fit. Only  $\sigma_L$  and  $\kappa$  show substantial contraction; the rate parameters  $\{p_{Am}\}$ ,  $[p_M]$ ,  $v$ , and  $[E_G]$  are largely prior-driven, consistent with Table 7.

overlaid on observed data (black points), propagating parameter uncertainty.

- **PPC plots** (`plot(bdeb_ppc(fit))`): Replicated data (grey) vs. observed data (red), assessing model adequacy (Gelman *et al.* 2013, Ch. 6).
- **Dose-response** (`plot_dose_response()`): Full-model predicted relative growth across concentrations with posterior uncertainty bands,  $EC_{50}$  and NEC marked.
- **Prior vs. posterior** (`type = "prior_posterior"`): Overlaid prior (red) and posterior (blue) densities for each parameter. This directly visualises the contraction (Section 6.2): parameters where red and blue overlap are prior-dominated; parameters where the posterior is much narrower are data-informed.

Figure 11 shows the prior-posterior comparison for the Neuhauser (1980) fit. The plot confirms the contraction analysis:  $\sigma_L$  is sharply data-determined,  $\kappa$  shows moderate updating, while  $\{p_{Am}\}$ ,  $[p_M]$ ,  $v$ , and  $[E_G]$  remain largely prior-shaped.

Figure 12 shows a prior predictive check.

## 6.6. Computational performance

Table 10 reports wall-clock times on a 16-core workstation (AMD Ryzen 9, 64 GB RAM, Ubuntu 22.04, CmdStan 2.36.0). Compilation is a one-time cost cached by **cmdstanr**; sampling times are for 4 chains  $\times$  2000 iterations.

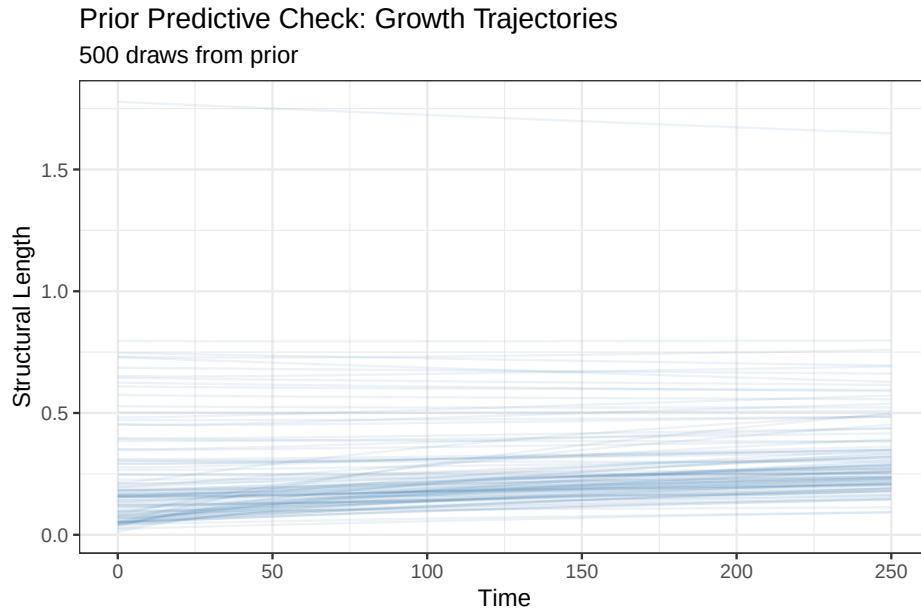


Figure 12: Prior predictive check: 500 growth trajectories sampled from the prior. The spread shows the range of data the model considers plausible *before* seeing data.

Table 10: Wall-clock times (seconds). Threading shows `parallel_chains`  $\times$  `threads/chain`.

| Model        | Data size         | Compile | Sample      | Threading    | Speedup      |
|--------------|-------------------|---------|-------------|--------------|--------------|
| Individual   | 1 ind., 13 obs.   | 30      | 69          | $4 \times 1$ | —            |
| Individual   | 1 ind., 37 obs.   | —       | 392         | $4 \times 1$ | —            |
| Hierarchical | 21 ind., 273 obs. | 35      | $\sim 3200$ | $4 \times 1$ | $1 \times$   |
| Hierarchical | 21 ind., 273 obs. | —       | $\sim 900$  | $2 \times 4$ | $3.6 \times$ |
| DEBtox       | 4 grp., 280 obs.  | 35      | $\sim 600$  | $2 \times 2$ | —            |

**Threading speedup.** Table 10 shows the hierarchical model under two threading configurations. With  $2 \times 4$  (2 parallel chains, 4 threads each, 8 cores total), wall-clock time is  $3.6 \times$  shorter than the sequential baseline ( $4 \times 1$ , 4 parallel chains, 1 thread each). Note that the two configurations use different total core counts (8 vs. 4), so this figure reflects both within-chain parallelism via `reduce_sum` and the change in parallel-chain scheduling; it is not a pure threading benchmark. The within-chain speedup arises because `reduce_sum` distributes the 21 per-individual ODE solves across threads within each chain.

**Scaling with dataset size.** Sampling time scales approximately linearly with the number of individuals in the hierarchical model, as each individual contributes an independent ODE solve distributed via `reduce_sum`.

**Memory usage.** Memory consumption scales linearly with  $N_{\text{ind}} \times N_{\text{obs}} \times N_{\text{states}}$ . For the bundled datasets, peak RSS was  $< 500$  MB per chain. The dominant memory cost is the ODE solver’s internal state, not the `Stan` parameter vector.

Table 11: Comparison of **BayesianDEB** with existing DEB calibration tools.

| Feature                  | AmP/ <b>DEBtool</b> | <b>deBInfer</b> | Custom Stan | <b>BayesianDEB</b>      |
|--------------------------|---------------------|-----------------|-------------|-------------------------|
| Bayesian inference       | —                   | Yes             | Yes         | Yes                     |
| Pre-built DEB models     | Yes                 | —               | —           | Yes                     |
| Hierarchical models      | —                   | —               | Manual      | Built-in                |
| DEBtox (TKTD)            | Partial             | —               | Manual      | Built-in                |
| Observation model choice | —                   | —               | Manual      | Switchable              |
| Prior specification API  | —                   | Basic           | Manual      | 6 families              |
| Temperature correction   | Manual              | —               | Manual      | Built-in                |
| Convergence diagnostics  | Minimal             | Basic           | Manual      | Automated               |
| PPC                      | —                   | —               | Manual      | Built-in                |
| LOO-CV                   | —                   | —               | Manual      | Built-in                |
| Within-chain parallelism | —                   | —               | Manual      | <code>reduce_sum</code> |
| R package                | MATLAB+R            | Yes             | —           | Yes                     |

## 7. Comparison with existing software

Table 11 summarises the feature landscape. **DEBtool\_M** and the AmP estimation tool (Marques *et al.* 2018) are the standard DEB calibration platforms but are primarily MATLAB-based, use deterministic optimisation, and do not support hierarchical models or formal uncertainty quantification. **deBInfer** (Boersch-Supan *et al.* 2017) provides Bayesian ODE inference in R but requires the user to write the ODE system and does not include DEB-specific models, ecotoxicological extensions, or Stan-based samplers. Custom Stan code can implement any of these models but requires significant expertise and has no reusable API. **BayesianDEB** combines the model coverage of AmP/**DEBtool** with the Bayesian inference capabilities of Stan, wrapped in a user-friendly R interface.

## 8. Current limitations and future work

- The "individual" and "growth\_repro" models support exactly one individual; multi-individual data require "hierarchical".
- The "debtox" model fits one ODE per concentration group (group-level); individual-level hierarchical DEBtox is planned.
- The hierarchical model places random effects on  $\{p_{Am}\}$  only.
- Temperature correction is global (single  $T$  per experiment); time-varying temperature is planned.
- The survival endpoint is not yet implemented.
- All lengths are structural ( $L = V^{1/3}$ ); the shape coefficient  $\delta_M$  is not estimated.
- R-side simulators (`deb_simulate`, `bdeb_prior_predictive`, trajectory plots) use the LSODA solver from **deSolve**, which automatically switches between stiff (BDF) and

non-stiff (Adams) methods. For the mildly stiff DEB system, LSODA predominantly selects BDF steps, matching the fixed-BDF solver in **Stan**; numerical agreement was verified to  $< 10^{-5}$  cm across 100 random parameter draws (Section 6.1).

- Strong posterior correlations between  $[p_M]$ ,  $v$ , and  $[E_G]$  (and between  $\{p_{Am}\}$  and  $[E_G]$ ) are inherent to the DEB  $\kappa$ -rule structure (Section 6.2). Informative priors are recommended.
- Comprehensive SBC validation (Section 6.1) has been performed for the individual growth model (397 valid replications). Preliminary SBC for the hierarchical model (38 valid replications) provides encouraging but statistically limited evidence. The DEBtox model has not yet been validated via SBC; it shares the same ODE system and observation likelihoods as the individual model, but its toxicokinetic damage component ( $D_w$ ,  $z_w$ ,  $b_w$ ) warrants dedicated validation. Comprehensive SBC for the hierarchical and DEBtox models is planned for a future version.
- Reparametrising the **Stan** models in terms of derived quantities ( $L_\infty$ ,  $g$ ,  $k_M$ ) rather than primary parameters ( $\{p_{Am}\}$ ,  $[p_M]$ ,  $v$ ) could improve both sampling efficiency and interpretability, as these quantities are better identified by growth data. This requires a Jacobian adjustment in **Stan** and modification of all four model programs.
- Multi-endpoint inference (growth + reproduction + respiration simultaneously) would substantially improve identifiability of rate parameters that are prior-dominated under growth-only data (Table 7). A joint likelihood combining Equations (11) and (12) with a respiration observation model is a natural extension.

## 9. Summary

**BayesianDEB** provides the first general-purpose R package for Bayesian Dynamic Energy Budget modelling. By embedding DEB ODEs within **Stan**'s HMC framework, it delivers full posterior uncertainty propagation to all parameters and derived quantities, principled prior incorporation from the AmP database, hierarchical partial pooling across individuals, and ecotoxicological  $EC_{50}$ /NEC estimation with posterior uncertainty. The declarative R interface requires no **Stan** programming from the user, while the within-chain parallelism via `reduce_sum` makes hierarchical and DEBtox models tractable on modern multi-core hardware. Equally important, the identifiability analysis and prior sensitivity tools make explicit a fundamental limitation that deterministic calibration tools obscure: for growth-only data, DEB rate parameters are largely prior-dominated, and species-specific informative priors or hierarchical data structures are essential for quantitative inference. We view this transparency as a feature, not a limitation, of the Bayesian approach.

## Computational details

The results in this paper were obtained using R 4.5.2 (2025-10-31) on x86\_64-pc-linux-gnu (Ubuntu 22.04) with the packages listed below. **BayesianDEB** 0.1.2 is distributed under the MIT license and available from the SCIOm r-universe at <https://sciom.r-universe.dev/>

**BayesianDEB** and from GitHub at <https://github.com/sciom/BayesianDEB>. **cmdstanr** is available from the Stan r-universe at <https://stan-dev.r-universe.dev>.

```
R> install.packages("BayesianDEB",
+   repos = "https://sciom.r-universe.dev")
R> install.packages("cmdstanr",
+   repos = "https://stan-dev.r-universe.dev")
R> cmdstanr::install_cmdstan()
```

R version 4.5.2 (2025-10-31), x86\_64-pc-linux-gnu

Attached packages:

BayesianDEB 0.1.2, posterior 1.6.1, bayesplot 1.14.0,  
ggplot2 4.0.1, deSolve 1.42, loo 2.8.0

Loaded via namespace:

cmdstanr 0.8.0 (CmdStan 2.36.0), rlang 1.1.7,  
cli 3.6.5, Rcpp 1.1.0

## Acknowledgments

This work was supported by the Croatian Science Foundation (HRZZ) under the project FUNECOSoil — Functional Ecological Characterization of Soils: The Basis for Ecotoxicological Classification (grant no. IP-2022-10-7177).

## References

- Betancourt M, Girolami M (2015). “Hamiltonian Monte Carlo for Hierarchical Models.” In SK Upadhyay, *et al.* (eds.), *Current Trends in Bayesian Methodology with Applications*, pp. 79–101. CRC Press, Boca Raton. doi:10.1201/b18502.
- Boersch-Supan PH, Ryan SJ, Johnson LR (2017). “**deBInfer**: Bayesian Inference for Dynamical Models of Biological Systems in R.” *Methods in Ecology and Evolution*, 8(4), 511–518. doi:10.1111/2041-210X.12679.
- Carpenter B, Gelman A, Hoffman MD, Lee D, Goodrich B, Betancourt M, Brubaker MA, Guo J, Li P, Riddell A (2017). “Stan: A Probabilistic Programming Language.” *Journal of Statistical Software*, 76(1), 1–32. doi:10.18637/jss.v076.i01.
- Gelman A (2006). “Prior Distributions for Variance Parameters in Hierarchical Models.” *Bayesian Analysis*, 1(3), 515–534. doi:10.1214/06-BA117A.
- Gelman A, Carlin JB, Stern HS, Dunson DB, Vehtari A, Rubin DB (2013). *Bayesian Data Analysis*. 3rd edition. Chapman & Hall/CRC, Boca Raton. doi:10.1201/b16018.

- Gestel CAMV, Dis WAV, Breemen EMDV, Sparenburg PM, Baerselman R (1991). “Influence of Cadmium, Copper, and Pentachlorophenol on Growth and Sexual Development of *Eisenia andrei* (Oligochaeta; Annelida).” *Biology and Fertility of Soils*, **12**, 117–121. doi:10.1007/BF00341486.
- Hoffman MD, Gelman A (2014). “The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo.” *Journal of Machine Learning Research*, **15**(47), 1593–1623. URL <http://jmlr.org/papers/v15/hoffman14a.html>.
- Jager T, Heugens EHW, Kooijman SALM (2006). “Making Sense of Ecotoxicological Test Results: Towards Application of Process-Based Models.” *Ecotoxicology*, **15**(3), 305–314. doi:10.1007/s10646-006-0060-x.
- Jager T, Zimmer EI (2012). “Simplified Dynamic Energy Budget Model for Analysing Ecotoxicity Data.” *Ecological Modelling*, **225**, 74–81. doi:10.1016/j.ecolmodel.2011.11.012.
- Johnson LR, Ben-Horin T, Lafferty KD, McNally A, Mordecai E, Paaijmans KP, Pawar S, Ryan SJ (2015). “Understanding Uncertainty in Temperature Effects on Vector-Borne Disease: A Bayesian Approach.” *Ecology*, **96**(1), 203–213. doi:10.1890/13-1964.1.
- Kooijman SALM (2010). *Dynamic Energy Budget Theory for Metabolic Organisation*. 3rd edition. Cambridge University Press. doi:10.1017/CB09780511805400.
- Marques GM, Augustine S, Lika K, Pecquerie L, Domingos T, Kooijman SALM (2018). “The AmP Project: Comparing Species on the Basis of Dynamic Energy Budget Parameters.” *PLOS Computational Biology*, **14**(5), e1006100. doi:10.1371/journal.pcbi.1006100.
- Mathieu J (2018). “EGrowth: A Global Database on Intraspecific Body Growth Variability in Earthworm.” *Soil Biology and Biochemistry*, **122**, 71–80. doi:10.1016/j.soilbio.2018.04.004.
- Neuhauser EF, Hartenstein R, Kaplan DL (1980). “Growth of the Earthworm *Eisenia Foetida* in Relation to Population Density and Food Rationing.” *Oikos*, **35**, 93–98. doi:10.2307/3544730.
- Talts S, Betancourt M, Simpson D, Vehtari A, Gelman A (2018). “Validating Bayesian Inference Algorithms with Simulation-Based Calibration.” doi:10.48550/arXiv.1804.06788. 1804.06788.
- Vehtari A, Gelman A, Gabry J (2017). “Practical Bayesian Model Evaluation Using Leave-One-Out Cross-Validation and WAIC.” *Statistics and Computing*, **27**(5), 1413–1432. doi:10.1007/s11222-016-9696-4.
- Vehtari A, Gelman A, Simpson D, Carpenter B, Bürkner PC (2021). “Rank-Normalization, Folding, and Localization: An Improved  $\hat{R}$  for Assessing Convergence of MCMC.” *Bayesian Analysis*, **16**(2), 667–718. doi:10.1214/20-BA1221.

**Affiliation:**

Branimir K. Hackenberger  
Department of Biology  
Josip Juraj Strossmayer University of Osijek  
Osijek, Croatia  
and  
SCIOM d.o.o., Osijek, Croatia  
E-mail: [branimir@sciom.hr](mailto:branimir@sciom.hr)  
URL: <https://github.com/sciom/BayesianDEB>

Tamara Djerdj  
Department of Biology  
Josip Juraj Strossmayer University of Osijek  
Osijek, Croatia

Domagoj K. Hackenberger  
Ruđer Bošković Institute  
Zagreb, Croatia